Labour Supply 2: Labour Supply and Public Policy

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Discuss various public policies that affect labour supply
  - Taxes/subsidies, transfers, employment insurance, worker compensation
- Show effect of these policies on work decisions using economic models
- Examine statistical evidence for the model predictions

Income Maintenance Programs

Overview of Income Maintenance Programs

- Represent about 10% of GDP in Canada
- Primary aims:
  - Raise income of specific groups (parents, elderly)
  - Supplement low wages
- Major programs: child benefits, employment insurance, social assistance, pensions

Key Challenge

Balancing income support with work incentives

Government Transfers in Canada

In 2017, 83% of families received some form of government transfer:

Program	% Receiving	Average Amount
Federal Child Benefits	20.2%	$6,314
Employment Insurance	14.0%	$8,352
Social Assistance	11.3%	$8,367
Old Age Security	25.1%	$8,566
CPP/QPP	32.4%	$10,092

Issues in Program Design

Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

Work Disincentives

- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

Permanent vs. Transitory Income Loss

- Different situations require different program designs
- Temporary layoff → short-term support
- Long-term disability → longer support + reintegration costs

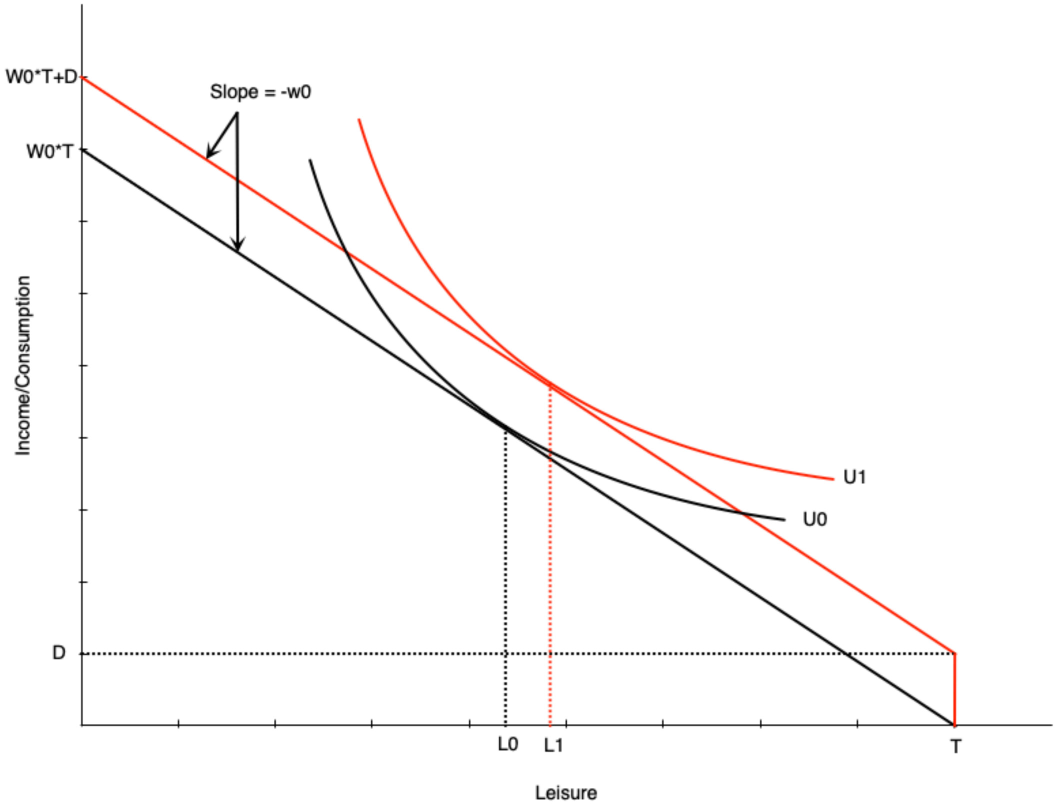
Demograns, Welfare, Wage Subsidies

Four Standard Income Support Programs

We will examine four policy approaches:

1. Demogrants - Fixed payment to all regardless of income/employment
2. Welfare - Lump-sum payment for non-workers, reduced dollar-for-dollar with earnings
3. Negative Income Tax (NIT) - Welfare payment that decreases at rate < 100% with earnings
4. Wage Subsidies - Addition to the wage rate

Demogrant: Theory



How it works:

- Unconditional lump sum transfer D
- Shifts budget line up parallel
- Wage unchanged → pure income effect

Labor supply impact:

- If leisure is normal → optimal leisure rises
- Reduces hours worked
- May induce non-participation

⚠ Work Incentive Effect

Unambiguously negative

Demogrant: Canadian Example

Universal Child Care Benefit (UCCB)

- Paid to families with children under 18
  - $160/month per child under 6
  - $60/month per child aged 6-17
- Replaced in 2016 by Canada Child Benefit (CCB)
  - More targeted to low/middle income families

Evidence (Schirle, 2015)

UCCB reduced labour supply of women with children under 6 by ~3.2 percentage points for low-educated married women

Welfare: Overview

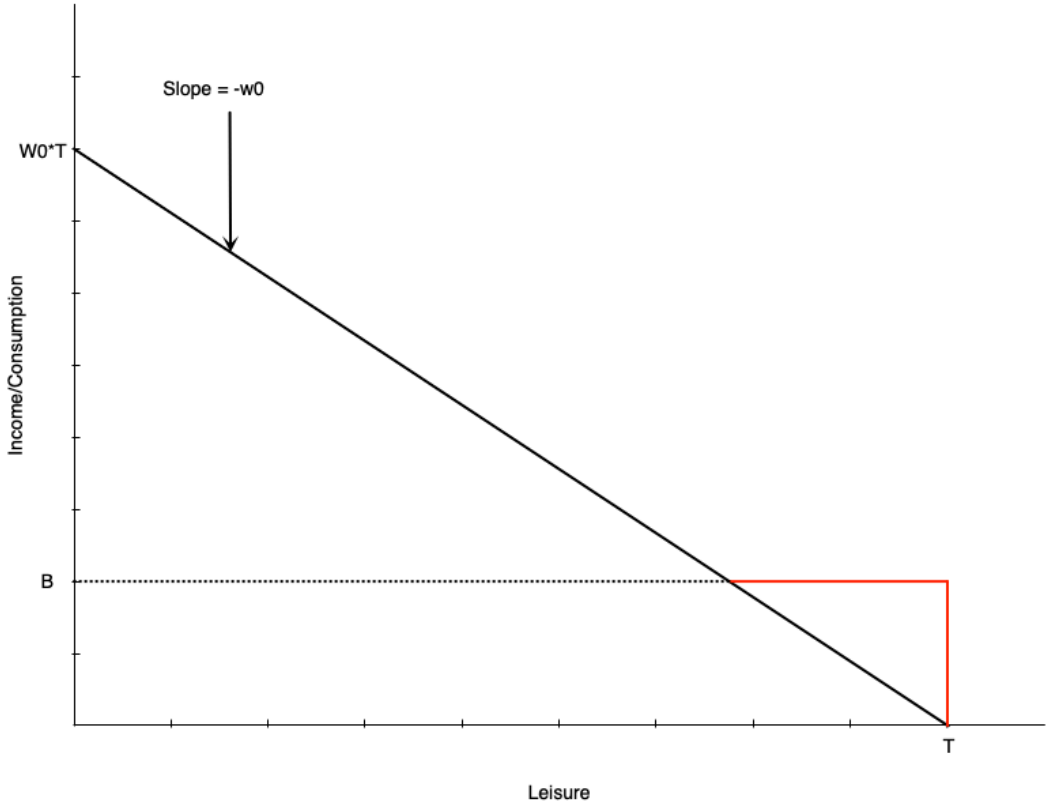
Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced dollar-for-dollar with earned income

Historical Criticism

Creates negative work incentives for some, but must balance against need for income support

Welfare: Budget Constraint



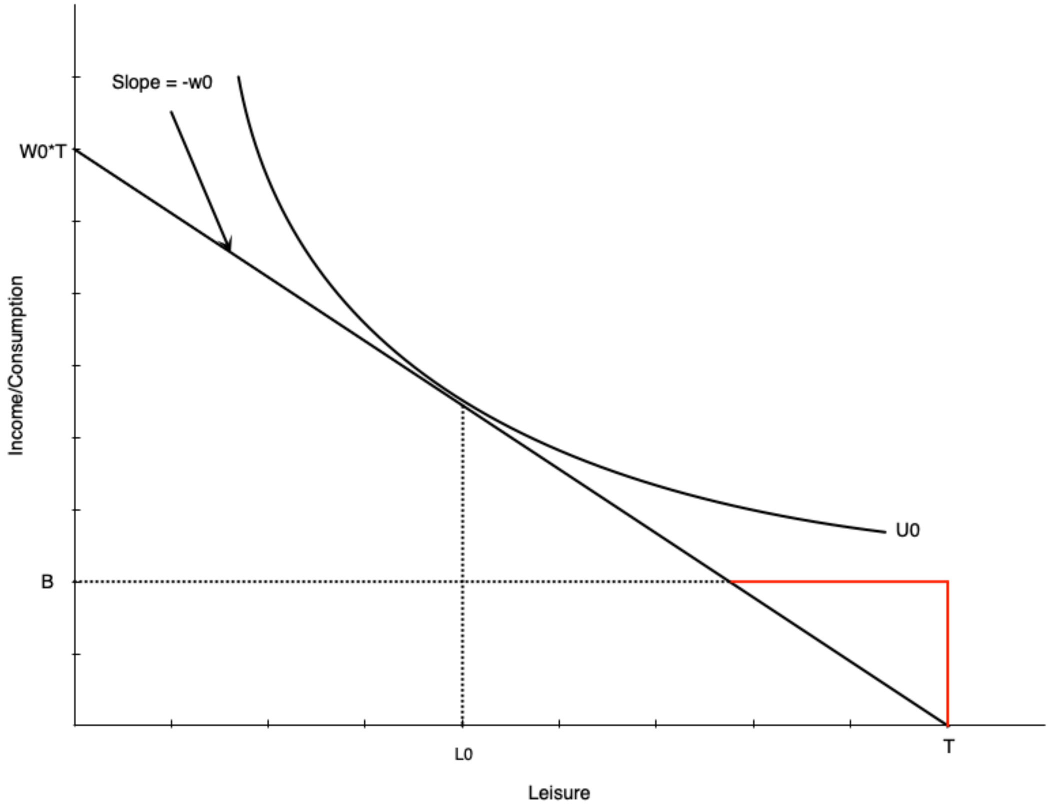
Budget line structure:

- Black line: original budget (leisure at L_0)
- Red then black: with welfare benefit B

Key features:

- Benefit B at zero hours of work
- Reduced $1 for each $1 earned
  - Equivalent to 100% tax rate on earnings
- Income constant at B until earned income equals benefit
- After that, earn wage w_0 per hour worked

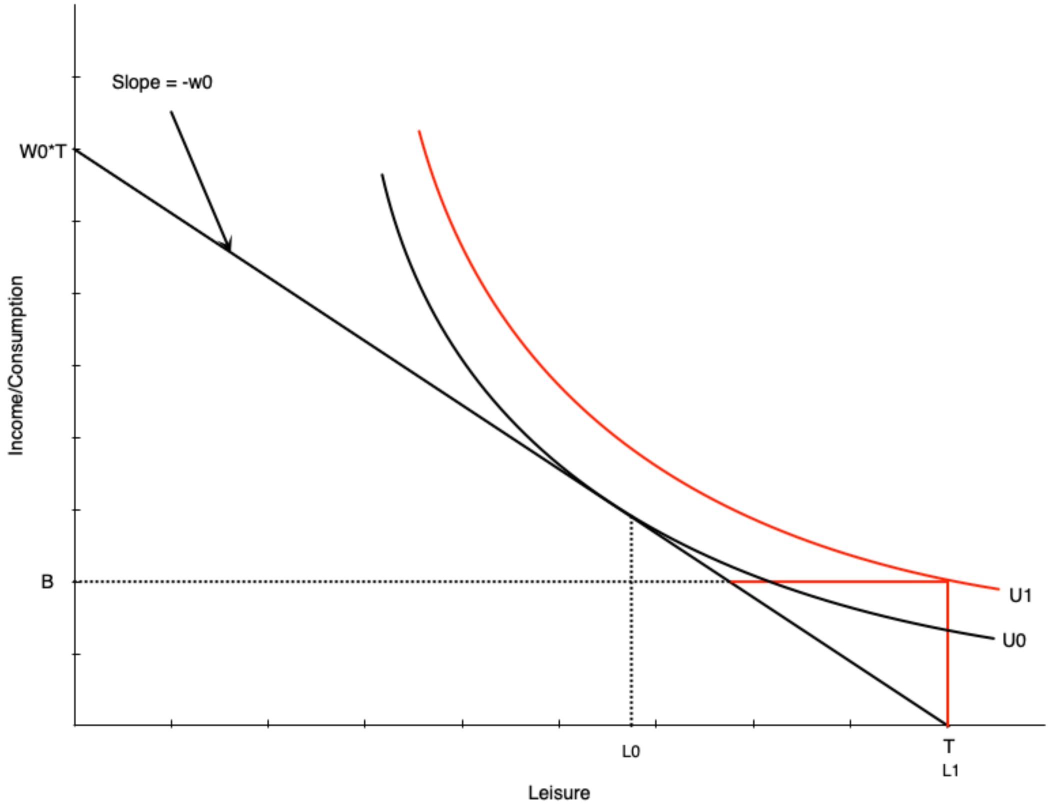
Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working H_0 hours before benefit
- Still maximizes utility at H_0 hours after
- Generally occurs with people already working many hours

Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses zero hours after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

Warning

This is the problematic effect of welfare programs

Combating Work Disincentives

Policy options:

1. Reduce benefit level
   - But may not provide adequate support
2. Increase wages
   - Through minimum wages, subsidies, training, unions
   - Can be expensive and may reduce labour demand
3. Change preferences
   - Alter attitudes toward work/leisure
   - Difficult to achieve
4. Negative income tax
   - Reduce tax-back rate on earnings
   - Discussed next…

Welfare: Evidence from Quebec

Natural experiment: Welfare benefit structure changed in 1989

Before 1989:

- Age-based benefits for singles/couples with no kids
- Under 30: $185/month
- 30 and over: $507/month

Results at age 30:

- Much longer time on welfare
- Lower employment rates

Note

Turning 30 created a large discontinuous jump in benefits

Negative Income Tax: Overview

What is NIT?

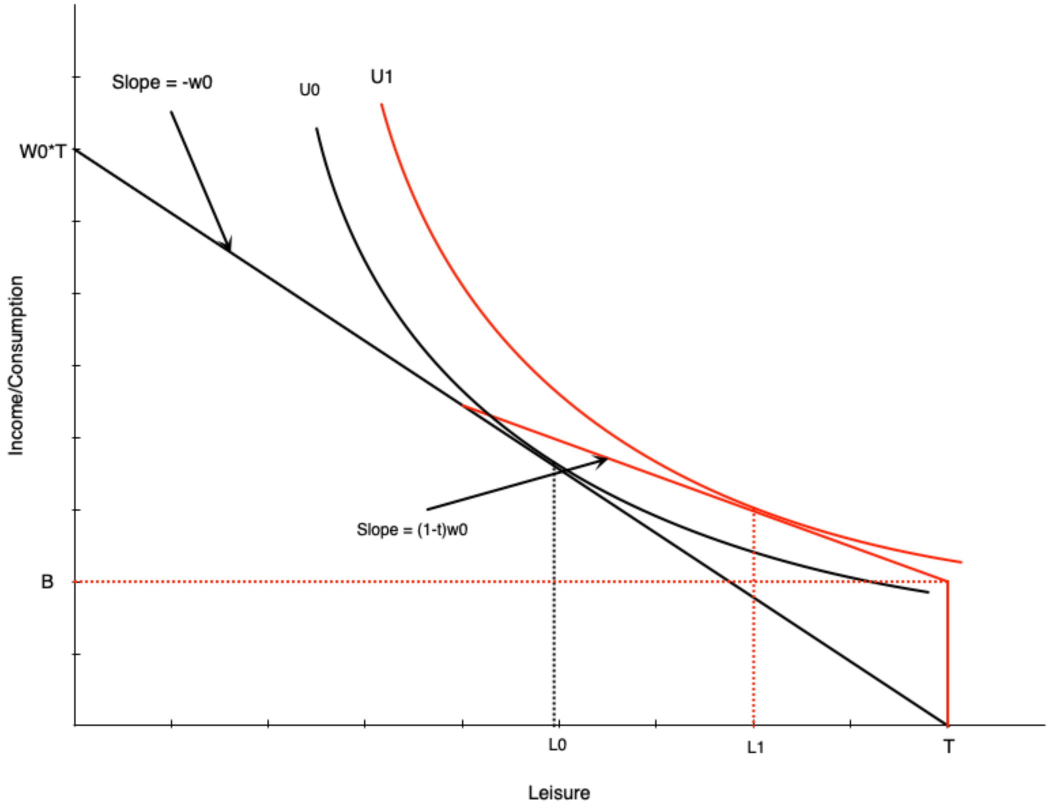
- Welfare program where benefit reduced at rate $< 100\%$ of earnings
- Example: $10,000 benefit reduced by 50¢ per dollar earned

Advantages

Provides income support with fewer work disincentives. Still maintains incentive to work since benefit reduces slowly.

Also known as: Guaranteed Annual Income (GAI) or Basic Income

Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit B at zero hours of work
- Tax rate t on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from H_0 to H_1
- Person still works some hours

Important

Work disincentive less than pure welfare

Negative Income Tax: Income and Substitution Effects

Income effects:

- Benefit increases non-labour income → increases leisure ↑
- Tax has negative income effect → reduces leisure ↓
- Net: benefit effect dominates → leisure ↑

Substitution effect:

- Tax rate reduces effective wage
- Makes leisure relatively cheaper → reduces work hours ↓

> **Overall Effect**
>
> Negative effect on work hours, but smaller than pure welfare

Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

Interpretation:

- Consumption = earned income (wh) + net benefit ($B - twh$) if net benefit > 0
- Otherwise just earned income (wh)

Break-even point: benefit reaches zero when $wh = \frac{B}{t}$

MINCOME: Manitoba's NIT Experiment

Program details (1970s):

- Randomized controlled trial
  - Treatment group received NIT
  - Control group did not
- Conducted in Dauphin (saturated - everyone got NIT) and Winnipeg
- 7 treatment groups with different benefit levels and tax rates
- Ended in 1979 due to political changes

MINCOME Treatment Groups

Guarantee at enrolment	Tax Rate (on total Income)		
	35%	50%	75%
$3800	Plan 1	Plan 3	*
$4800	Plan 2	Plan 4	Plan 7
$5480	X	Plan 5	Plan 8

Plan 9

* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.
X This option was deemed to expensive for the budget.
Plan 9 was the control group

Source: Wikipedia

MINCOME: Results

Labour supply effects:

- Small decline in hours worked
  - Mainly mothers of newborns
  - Women in two-parent families
  - No significant decline for men

Health outcomes:

- Fewer hospitalizations and doctor visits
- Especially for mental health and accidents

Education:

- Increase in adolescents finishing high school

Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: $5/hour subsidy increases $15/hour wage to $20/hour

Effect on budget constraint:

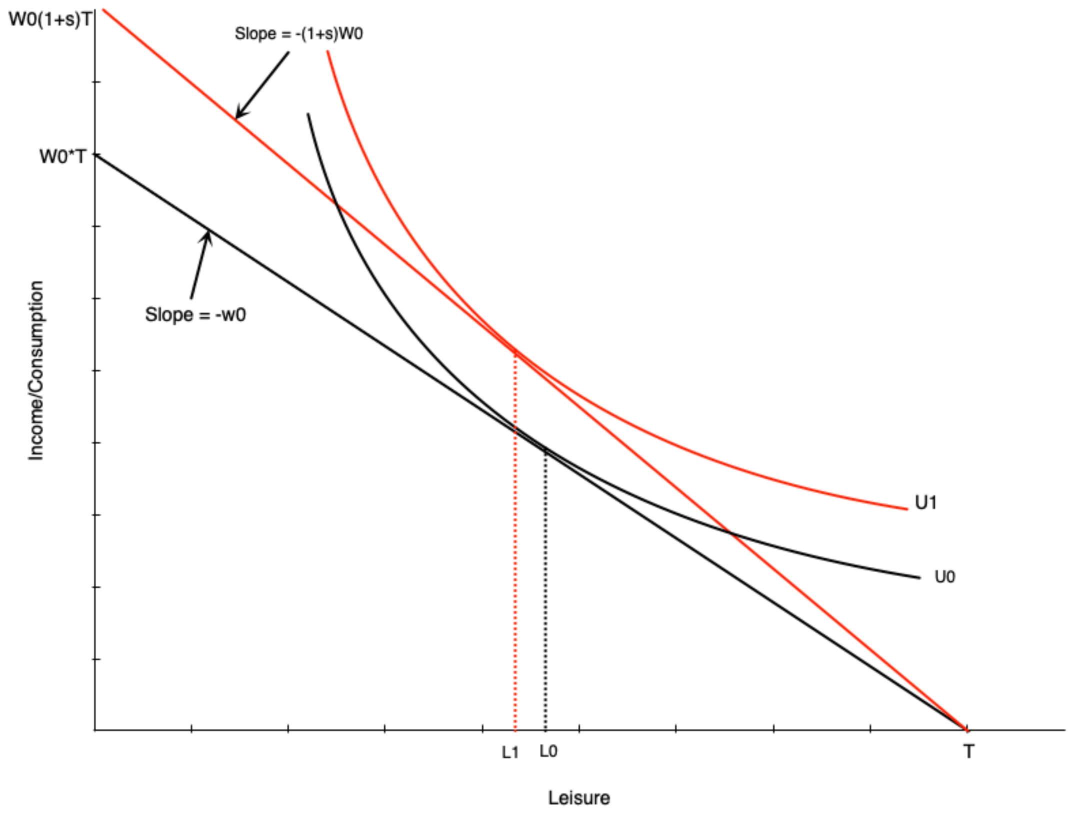
- Budget line pivots upward

Labour supply effects:

- Income effect → increases leisure ↑
- Substitution effect → decreases leisure ↓

> ⚠ **Warning**
>
> Overall effect on work hours is ambiguous

Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

Wage Subsidies: Policy Implications

Advantages

- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

Disadvantages

- Do not provide income support to non-workers
- May not help those in most need

In Canada:

- Pure wage subsidies mostly go to employers
- Unclear if any benefits passed on to workers

Tax Credits: Overview

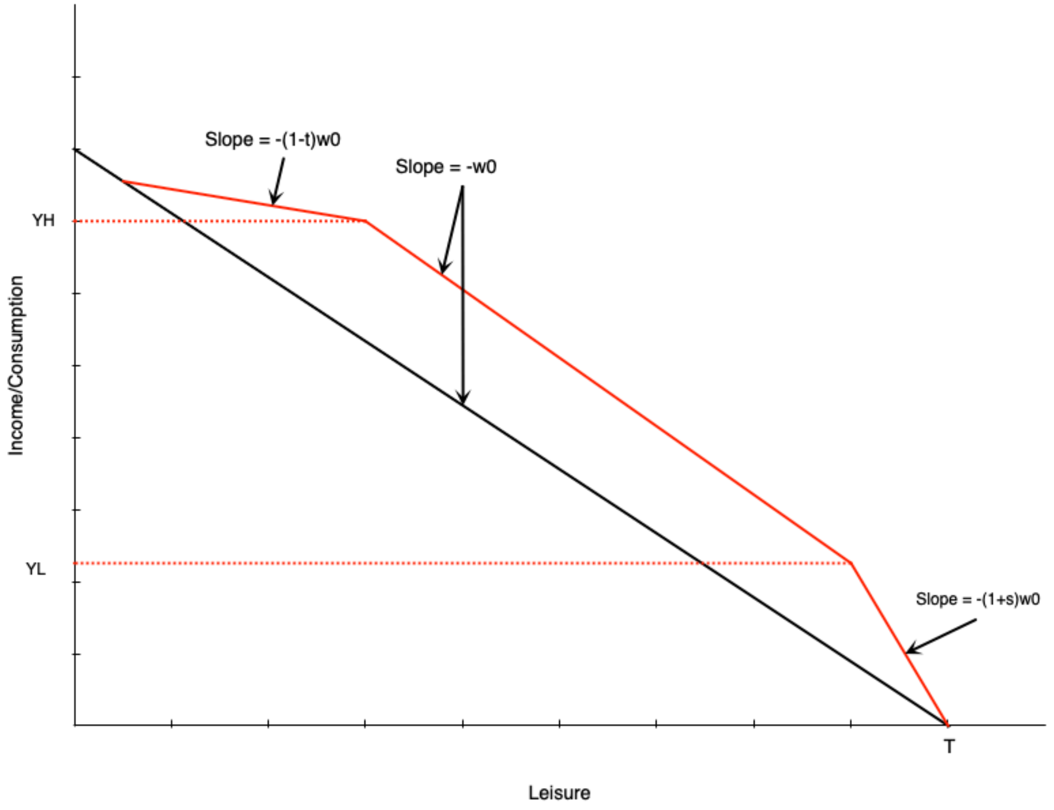
Two types of tax credits:

1. Non-refundable: Can reduce tax owed to zero (but not below)
2. Refundable: Can reduce tax owed below zero (results in payment)

Income Maintenance Programs via Refundable Tax Credits

- Canada Worker Benefit (CWB) - Canada
- Earned Income Tax Credit (EITC) - US

Tax Credits: Budget Constraint



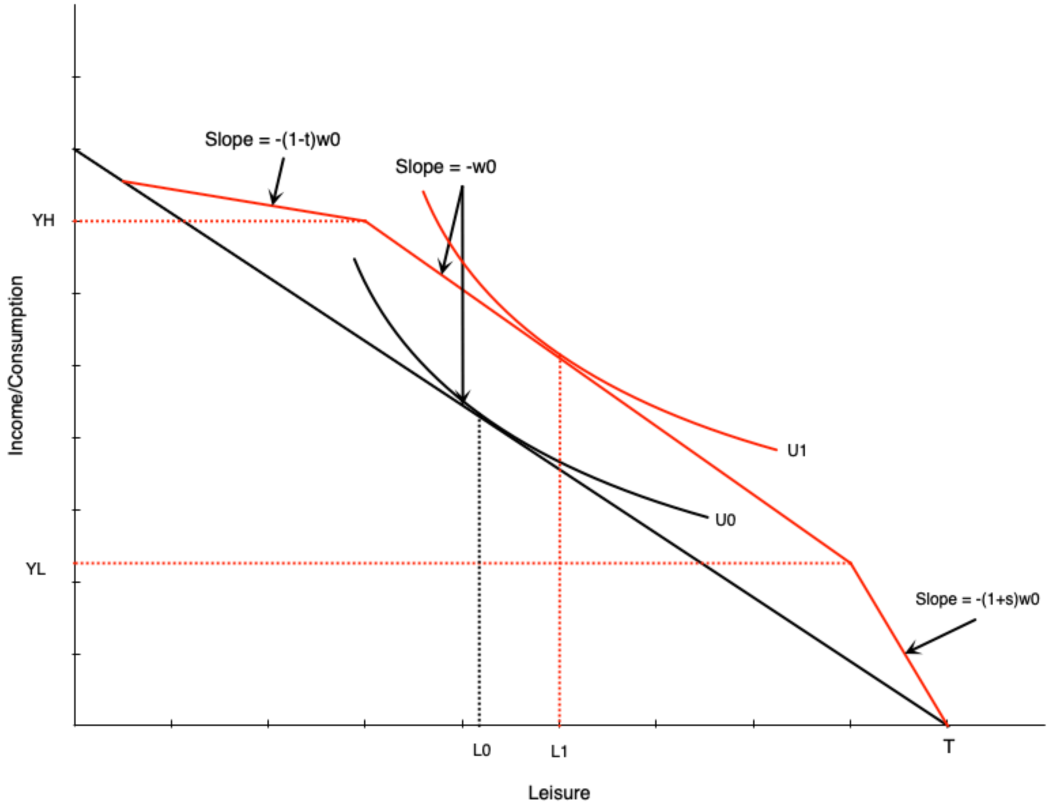
Three segments:

1. Phase-in: Subsidy proportional to earned income
2. Plateau: Maximum benefit level
3. Phase-out: Benefit reduced with earned income

Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

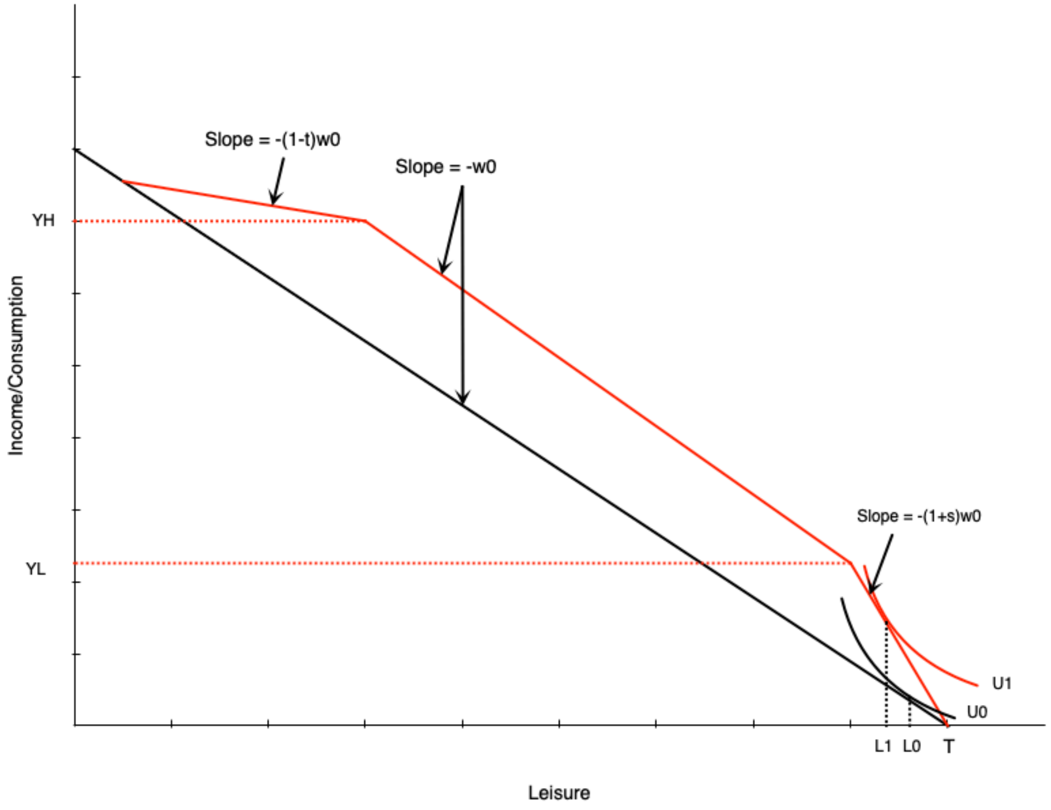
Tax Credits: Effect in Plateau Range



Person initially in middle range:

- Tax credit gives maximum benefit
- Acts like pure income effect
- If leisure is normal good → reduce work hours

Tax Credits: Effect in Phase-In Range



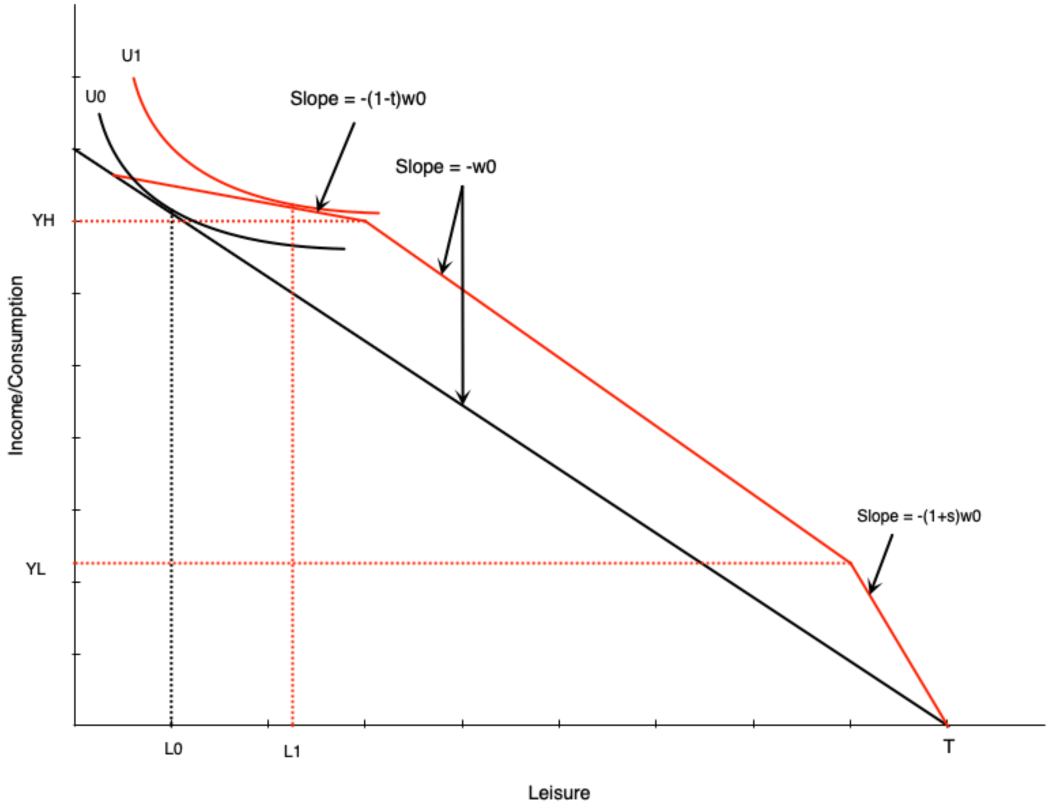
Person in phase-in range:

- Sees increase in effective wage
- Acts like pure wage subsidy
- Income and substitution effects in opposite directions

⚠ **Warning**

Effect on hours is ambiguous

Tax Credits: Effect in Phase-Out Range



Person in phase-out range:

- Similar to negative income tax
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

> **Important**
>
> Effect on work is unambiguously negative

Unemployment Insurance

Employment Insurance: Overview

Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

Funding:

- Payroll taxes on employers and employees

Two main components:

1. Regular benefits - for those unemployed through no fault of their own
2. Special benefits - sickness, maternity, parental leave, caregiving

Employment Insurance: Benefits

Regular Benefits Structure

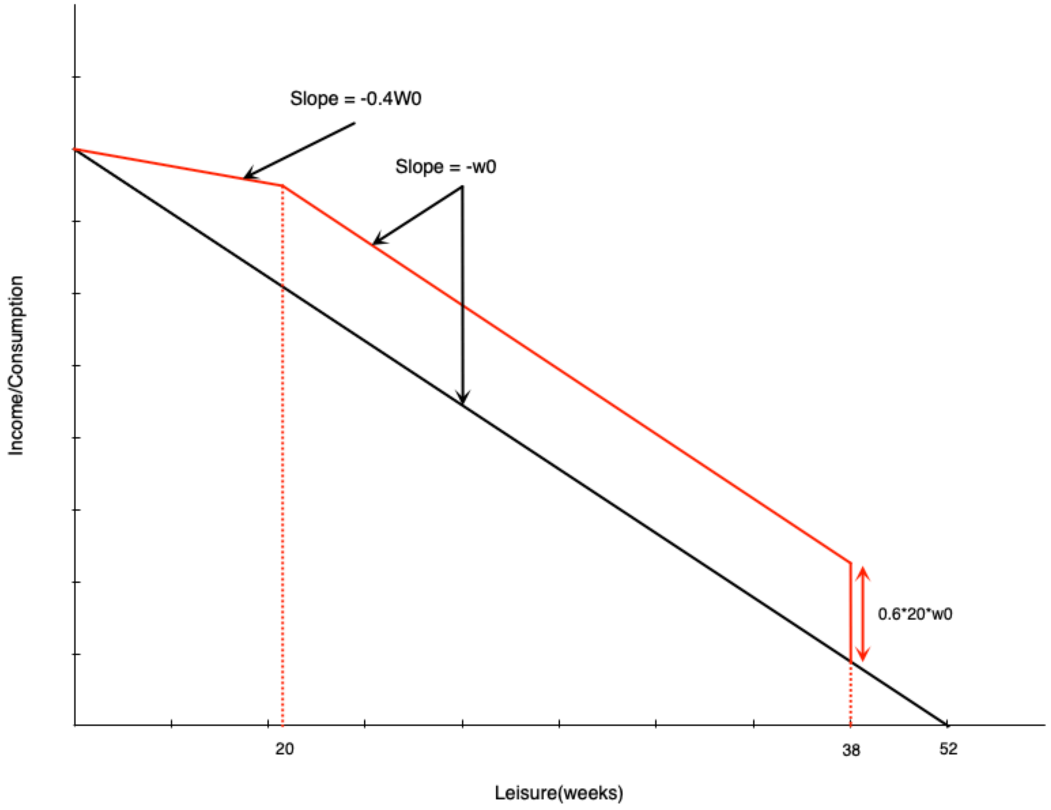
- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: 14 to 45 weeks
  - Depends on unemployment rate and hours worked in last year

EI: Simplified Model Parameters

To analyze with income-leisure diagram:

- Must work 14 weeks in the year to qualify
- Benefit is 60% of weekly wage
- Maximum of 20 weeks of benefits
- Can work up to 32 weeks and get full benefits
- Each week worked beyond 32 → lose benefits for that week
- 60% benefit = 60% tax on earnings beyond 32 weeks
- Cannot work and collect EI simultaneously

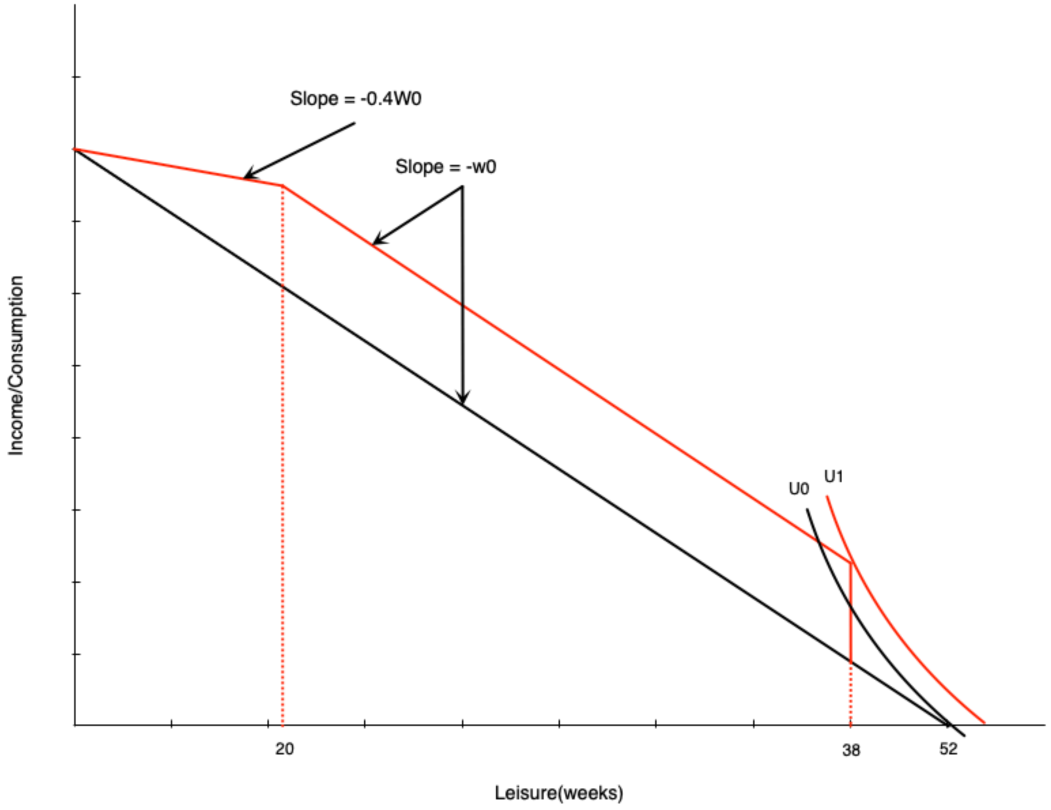
EI: Budget Constraint



Three segments:

1. First 14 weeks: Paid at usual wage w_0
2. Weeks 14-32 of work: Get 20 weeks of benefits
   - Benefit = $0.6 \times 20 \times w_0$
3. More than 32 weeks of work: Benefits reduced week-for-week
   - Benefits paid at 60% of wage
   - Work paid at 100% of wage
   - Keep 40% of wage in this range

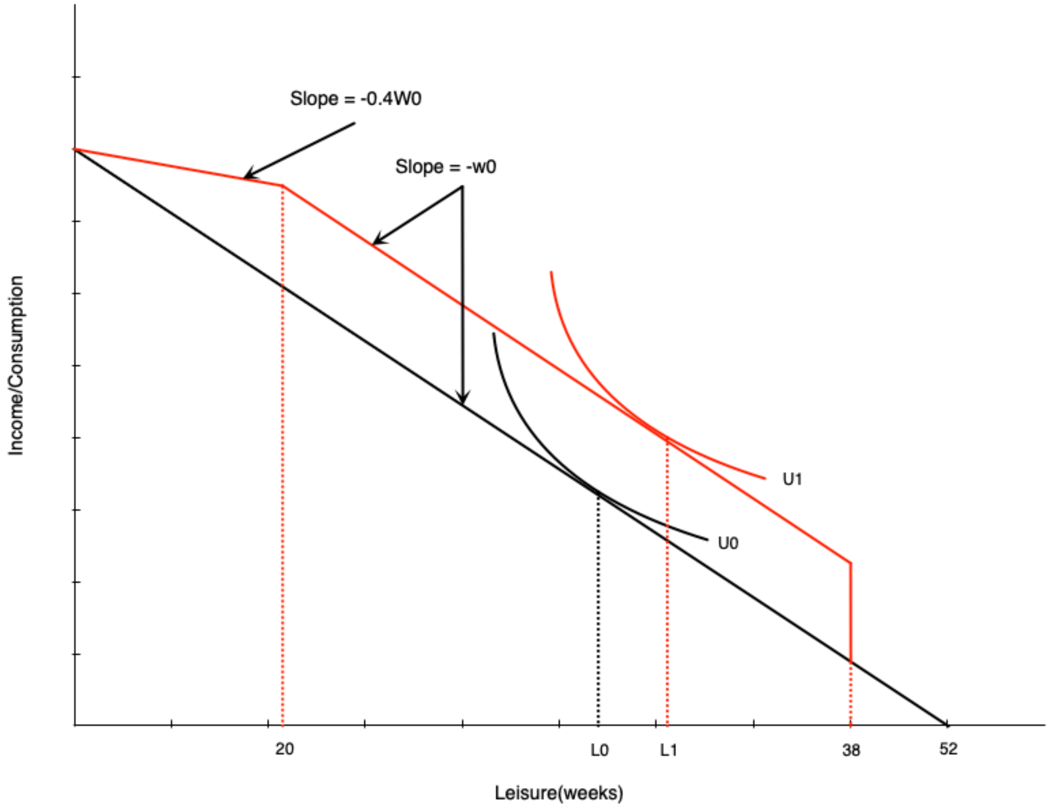
EI: Effect on Non-Participants



Effect on Non-Workers

A non-participant might start working due to large increase in non-labour income after 14 hours of work. Induces labour force entry.

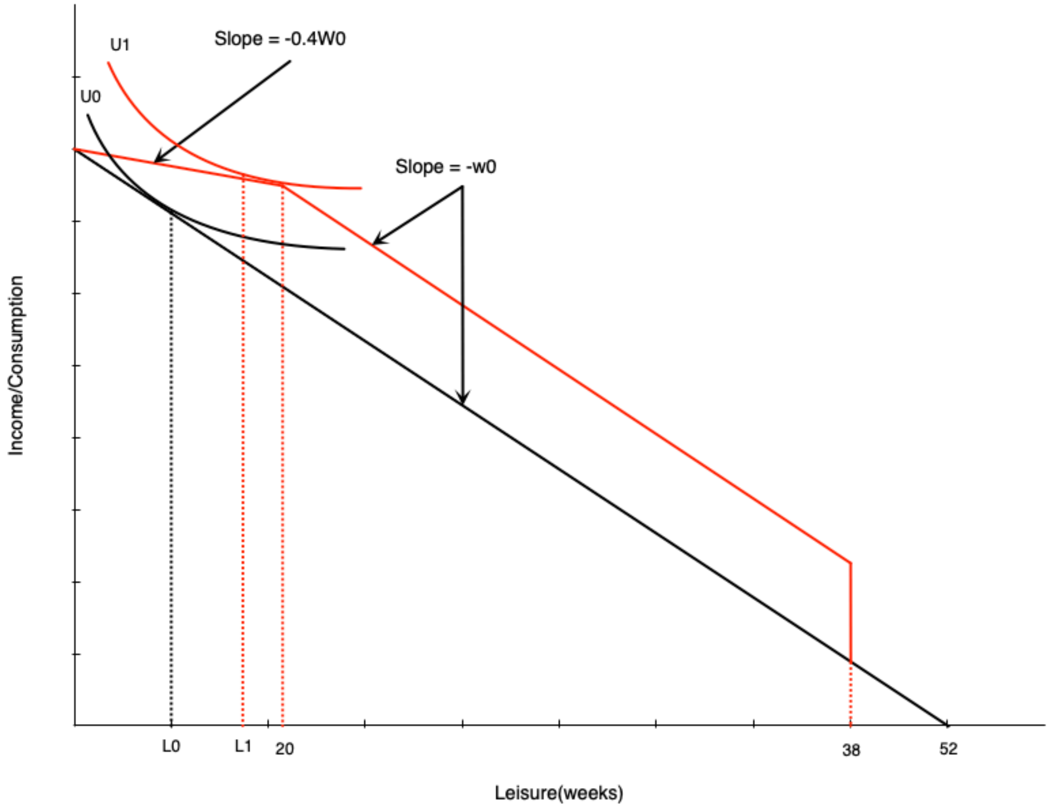
EI: Effect in Middle Range



Person getting full benefits:

- Acts like pure income effect
- Will reduce work weeks

EI: Effect on High-Hour Workers



In phase-out range:

- Acts like negative income tax
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

Important

Effect unambiguously negative

Evidence: Maine vs. New Brunswick

Natural experiment:

- Initially had very similar EI schemes
- Later, New Brunswick increased benefit generosity relative to Maine
- Difference-in-differences approach:
  - Compare changes in NB before/after to Maine over same period
  - Maine acts as control group
  - Controls for other time-varying factors

Evidence: Maine vs. New Brunswick Results

New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

Interpretation

- Corresponds exactly to model predictions
- Highlights work disincentive effects of income maintenance programs

Disability and Workers Compensation

Introduction

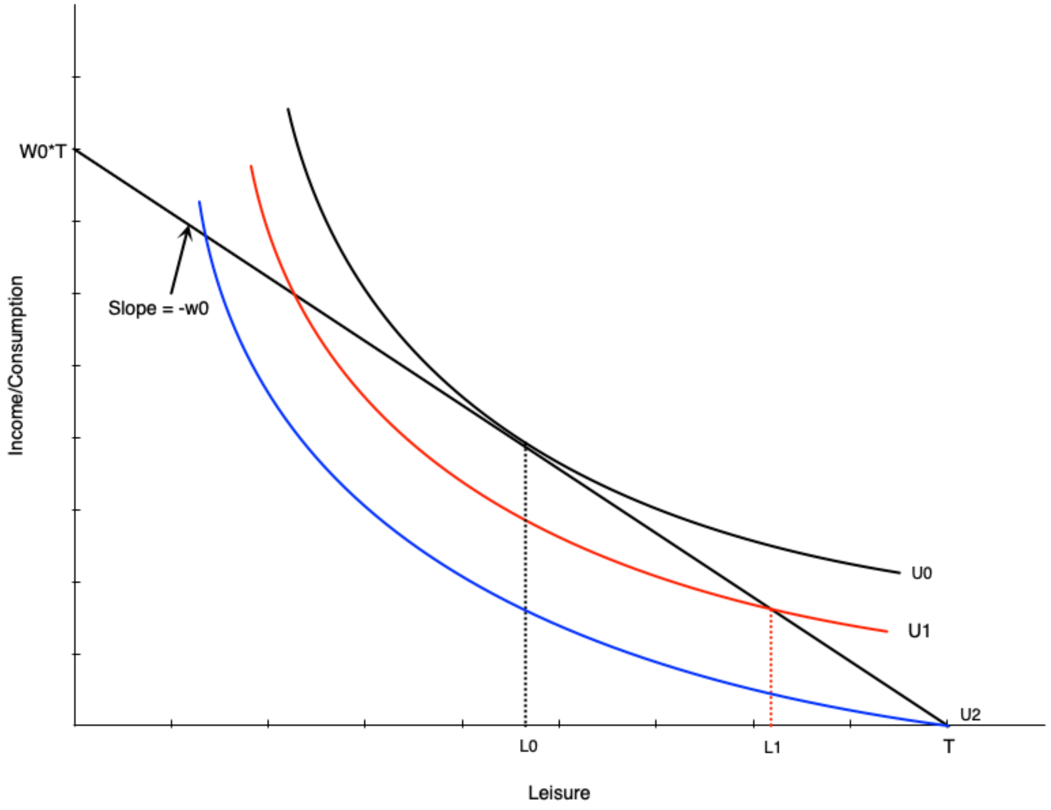
What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

Work Incentive Considerations

- Severe injury/disability: work incentives not a concern (unable to work)
- Partial disability: people may still be able to work

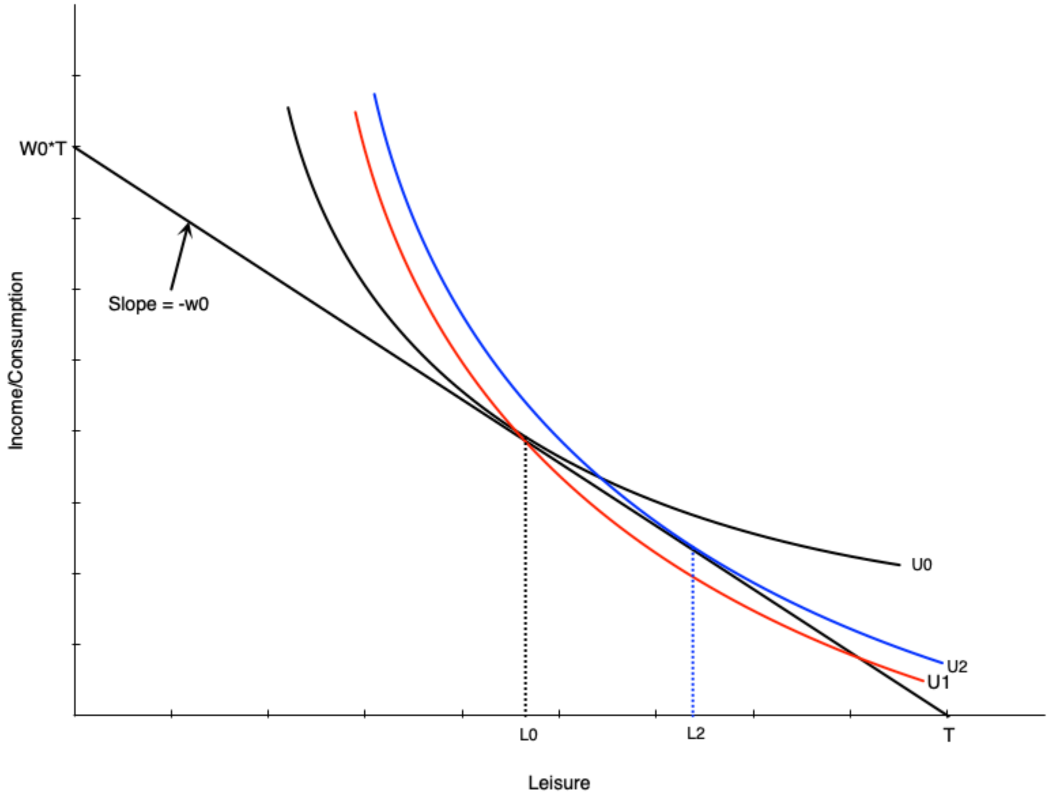
Effect of Disability on Labour Supply



Three scenarios:

1. Initially: Choose L_0 leisure, work H_0 hours
2. Injured but can work: Choose L_1 leisure, work H_1 hours
    - Lower (non-optimal) indifference curve
3. Completely disabled: Choose T leisure, zero work
    - Even lower indifference curve

Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

Result:

- Even if unconstrained, choose more leisure
- Person moves from L_0 to L_2

Compensation Schemes

Typical disability insurance structure:

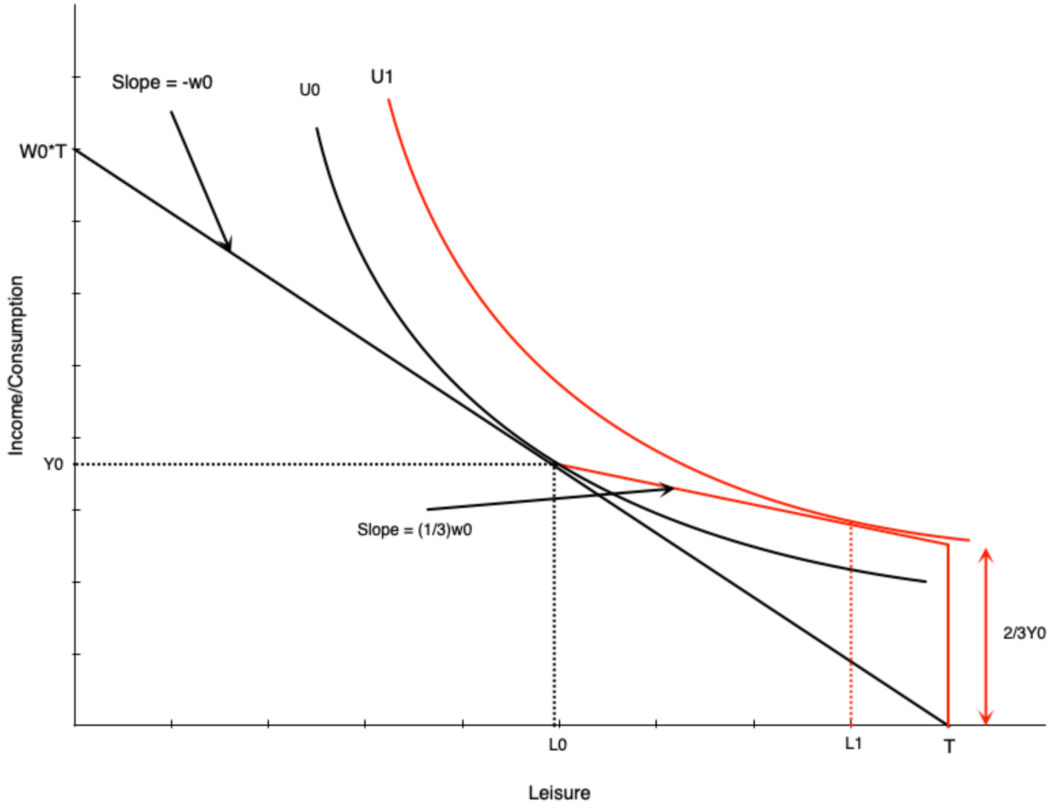
- Pays a fraction of previous earnings

> **Example: Ontario WSIB**
>
> - 85% of net earnings (up to maximum)
> - Lasts until return to work or age 65

Two schemes we'll examine:

1. Two-thirds income replacement
2. Full income replacement

Two-Thirds Compensation



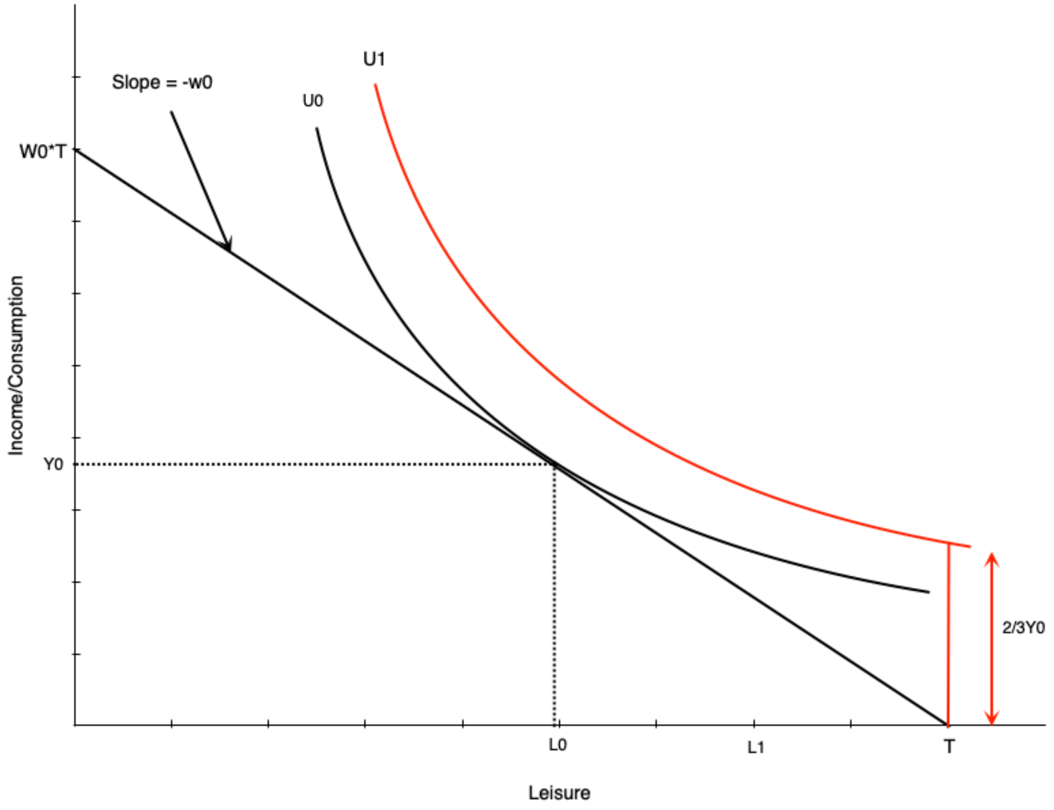
Looks like negative income tax:

- At zero hours: get 2/3 of previous income
- For each hour worked:
  - Gain full wage
  - Lose 2/3 of wage in benefits
  - Keep 1/3 of wage

For workers with flexible hours:

- Effects match NIT
- Negative effect on hours worked

Restricted Hours: Generous Compensation



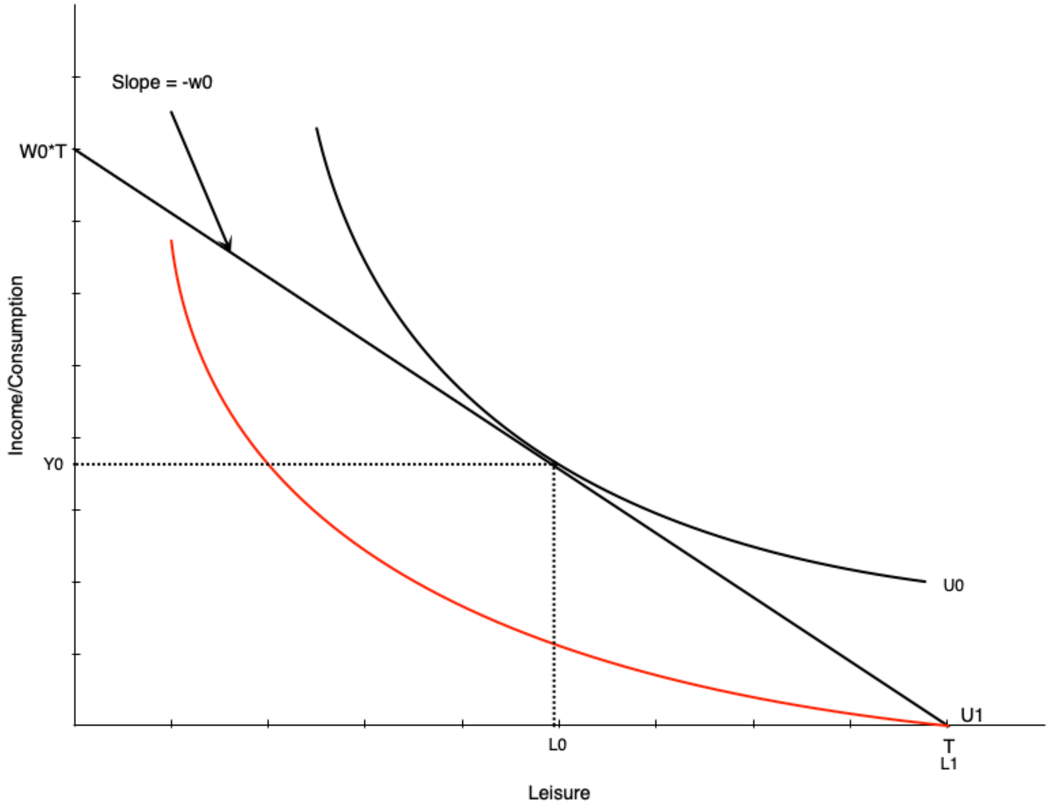
What if worker can only work H_0 hours or zero?

- If scheme generous enough → choose zero hours
- Utility at zero hours > utility at H_0 hours

Implication

Strong work disincentives of generous scheme

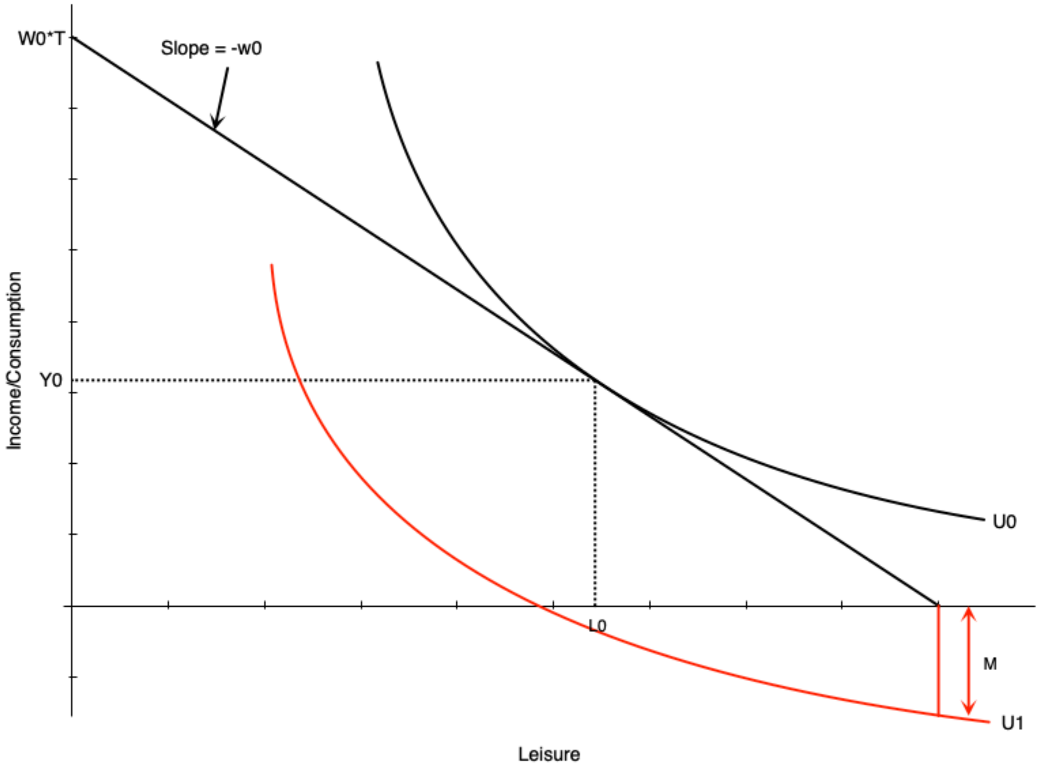
No Compensation Case



Not Providing Compensation is Problematic

- Person who can work chooses H_0 hours
- Person who cannot work gets lower utility
- End up substantially worse off

Medical Expenses Case

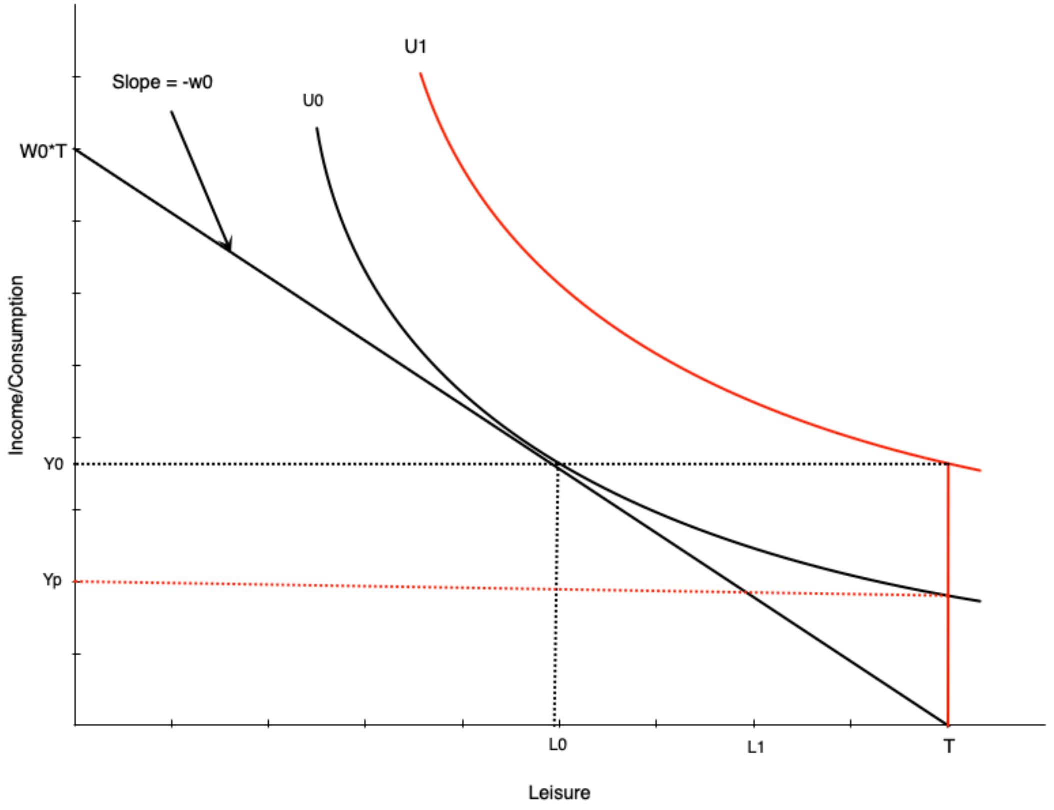


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

Optimal Compensation Level



How much to compensate non-workers?

Two options:

1. Full income replacement
   - Higher indifference curve (U_1) than working
2. Same satisfaction as working
   - Same indifference curve as working
   - Lower compensation (Y_p)

> **Note**
>
> The optimal level is a policy question

Child Care Subsidies

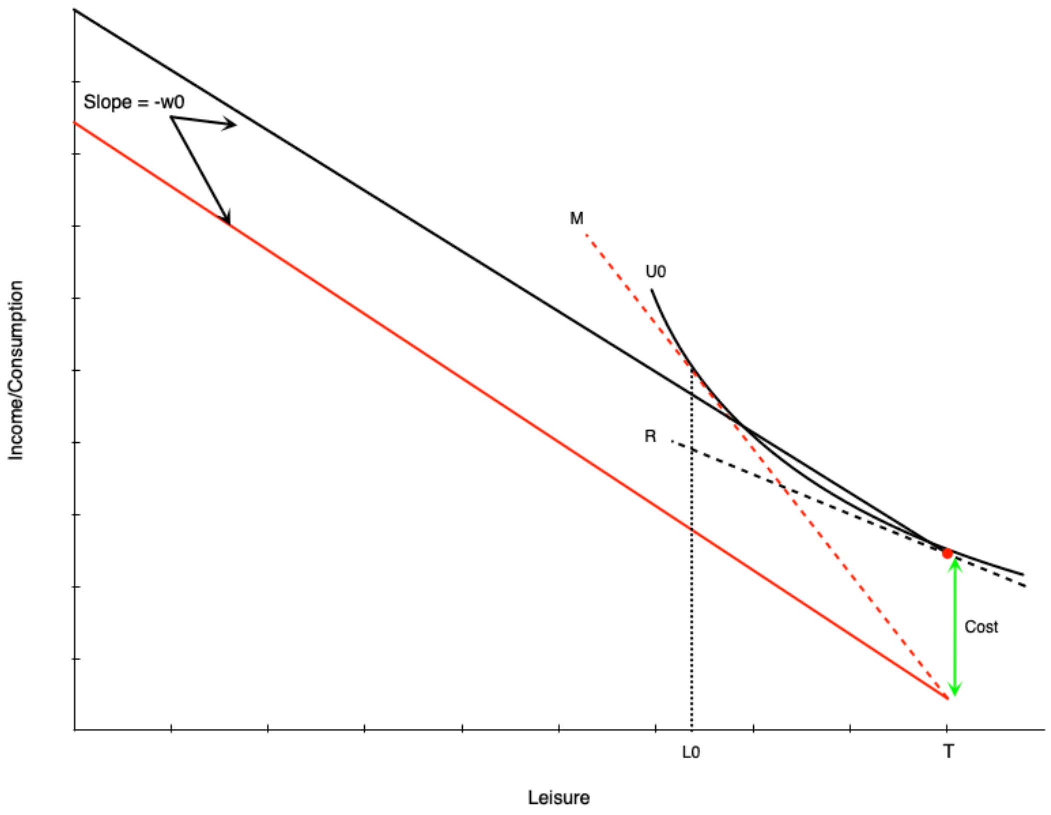
Introduction

Canada's child care expansion:

- Goal: reduce average cost to $10/day by 2026
- More than half of provinces already at $10/day average
- Subsidy varies substantially by province
- Requires provincial-federal agreements

> **Quebec Case Study**
>
> Had $10/day child care since 1997. Provides useful evidence on effects.

Child Care Costs: Non-Working Case



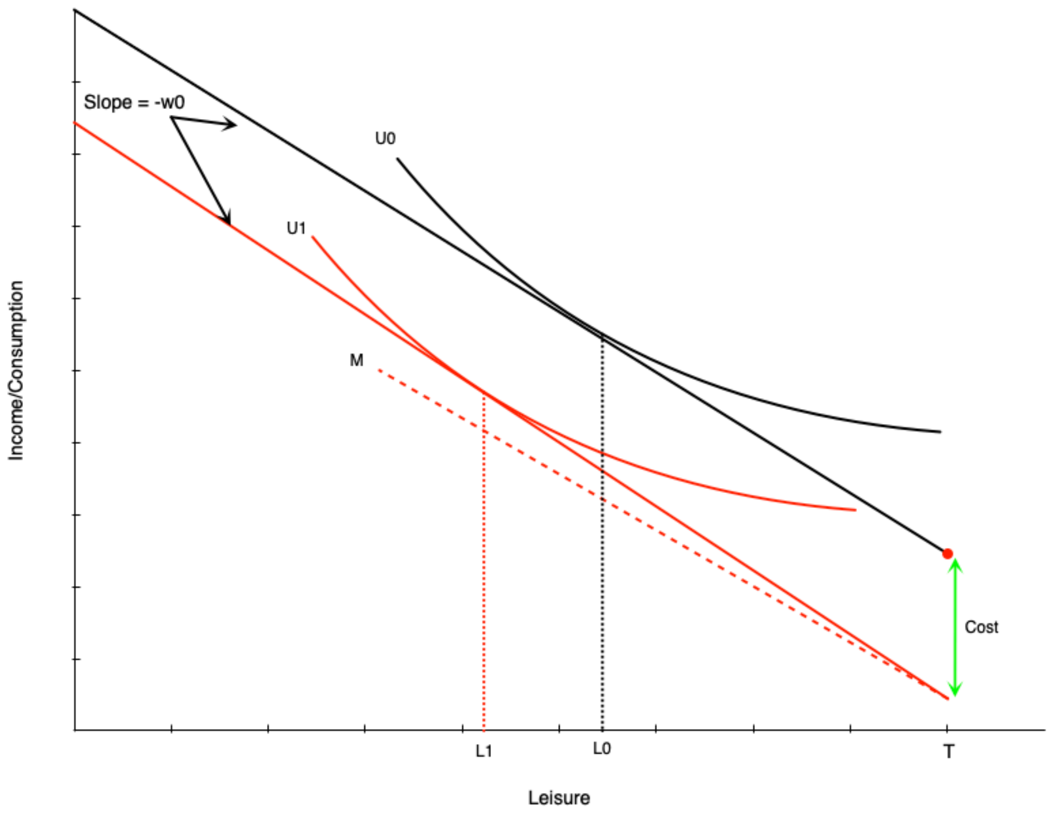
Budget line becomes discontinuous:

- At zero hours: red dot
- For any work: drops by child care costs
- Then straight line with slope = wage

Effect:

- Child care costs increase reservation wage
- Need higher wage to induce work
- This person chooses not to work

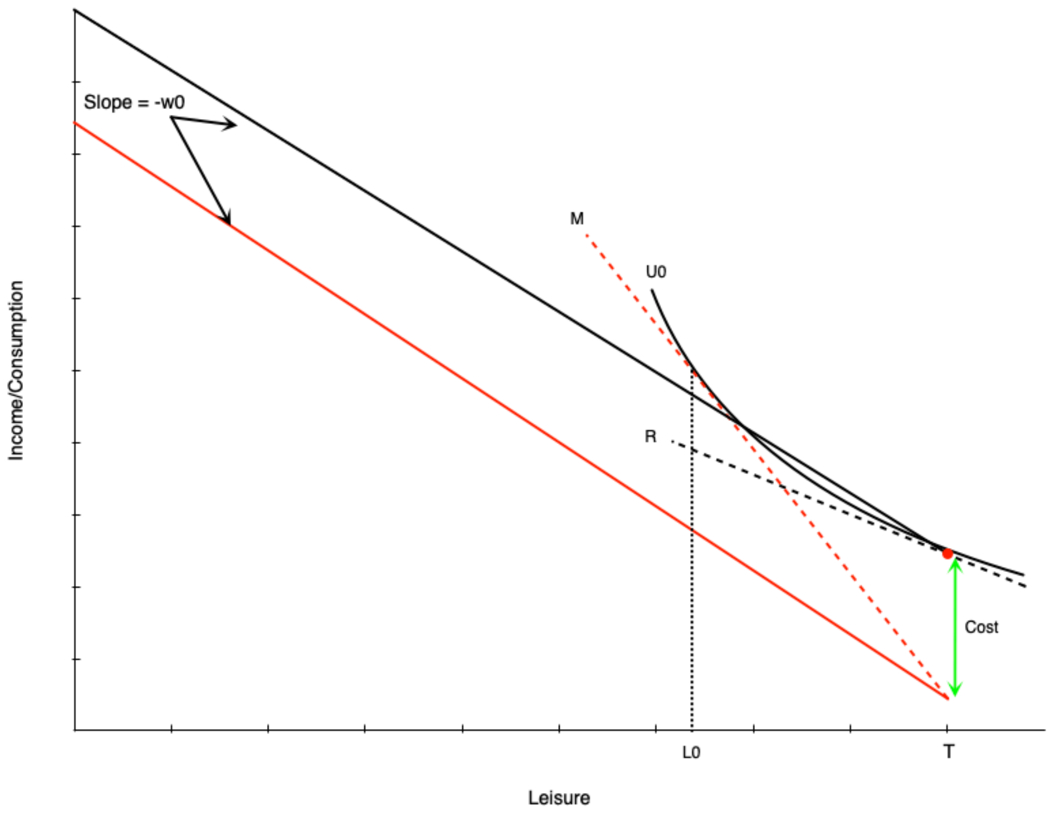
Child Care Costs: Working Case



Person with reservation wage above wage + costs:

- Works positive hours
- Child care costs = negative pure income effect
- Choose less leisure, more work
- Must work more to offset income loss

Child Care Costs: Labour Supply Gap



Creates Gap in Labour Supply Schedule

- Person who works will work minimum of $H_0 = T - L_0$ hours
- Not worth working less due to large child care costs
- Labour supply schedule jumps from zero to H_0 hours

Child Care Subsidies: Effects

A full subsidy (costs → zero) would:

1. Move budget line up in parallel
2. Reduce reservation wage
    - Lower barrier → encourages participation
    - Especially encourages part-time work
3. Create positive income effect
    - Increases consumption of leisure
    - Reduces work hours

Evidence: Quebec's $5/Day Program

Program details:

- Implemented in 2000
- Currently sliding scale: $8.25 to $21.45

Research design:

- Difference-in-differences approach
- Compared Quebec mothers to mothers in other provinces
- Other provinces = control group

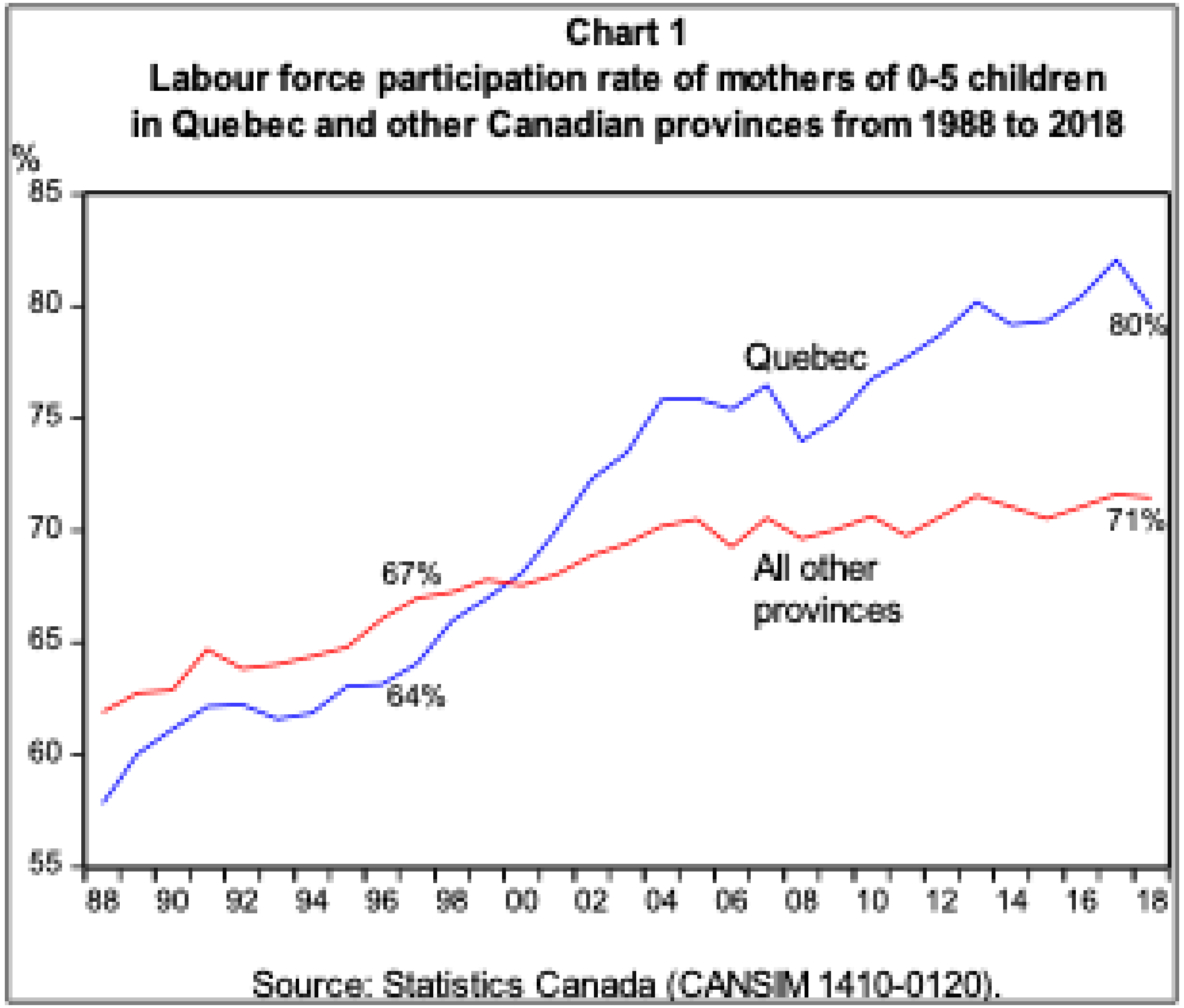
Key Finding

Labour force participation of mothers with young children ↑ 7.7 percentage points

Evidence: Quebec's $5/Day Program

> ⚠ **Warning**
>
> Unfortunately also found negative effects on child mental health, non-cognitive behaviour, crime later in life

Evidence: Visual Summary



Source: Fortin (2019)

Summary

Key Takeaways

Income maintenance programs:

- Provide crucial income support to various groups
- Can create work disincentives in some cases
- Must balance support needs against incentive effects

We examined several programs through the labour supply model:

- Demogrants, welfare, NIT, wage subsidies, tax credits
- Employment insurance
- Disability and workers compensation
- Child care subsidies